

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 07

Differentiation and Integration

1 Concepts | 42 questions

B5 Landscape | one question per teaching page

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Chapter 07 | Differentiation and Integration

CLASSROOM ROUTE

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)**42 questions**

Official lessons: 7-1, 7-2, 7-3, 7-4

The Formula Bank changes at each Concept boundary.

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

OFFICIAL LESSONS 7-1, 7-2, 7-3, 7-4

Fundamentals of Calculus (Limits and Derivatives)

Equation-first recall | use the same rail beside every question in this Concept

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Start with the family cue, write the first move, then use the working grid.

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F01 | V01

Q001 | WRITTEN

C07-K01-7-1-WU

For $f(x) = x^2 + 2x$, find the average rate of change from -1 to 3 , and from 3 to $3 + h$.

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F01 | V02

Q002 | WRITTEN

C07-K01-7-1-TRY

Find the average rate of change for the three given Try functions on their stated intervals.

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F01 | V03

Q003 | WRITTEN

C07-K01-7-1-EX

Find the average rate of change for all nine given Exercise functions, including fixed and h-form intervals.

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F01 | V04

Q004 | WRITTEN

C07-K01-7-1-CH

Find the average rate of change of a circle's area as its diameter changes from 10 cm to 18 cm.

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F02 | V01

Q005 | WRITTEN

C07-K01-7-2-WU

Find the **given** limits by direct substitution and by simplifying the indeterminate fraction before substitution.

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F02 | V02

Q006 | WRITTEN

C07-K01-7-2-TRY

Find the three given Try limits, including polynomial and h-quotient forms.

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F02 | V03

Q007 | WRITTEN

C07-K01-7-2-EX

Find all nine given Exercise limits, including x-limits and h-limits.

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F02 | V04

Q008 | WRITTEN

C07-K01-7-2-CH

If $\lim_{x \rightarrow 3} (x^2 - 5x + k)/(x - 3) = L$ is finite, find Lk .

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2} x^2 \quad A' = 3\sqrt{3} x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F03 | V01

Q009 | WRITTEN

C07-K01-7-3-WU

Find the differential coefficient of $f(x) = 3x^2$ at $x = -2$ using the definition.

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F03 | V02

Q010 | WRITTEN

C07-K01-7-3-TRY

Find the differential coefficients at the given values for the two given Try functions using the definition.

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F03 | V03

Q011 | WRITTEN

C07-K01-7-3-EX

Find the differential coefficients for all six given Exercise functions at their stated values.

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F03 | V04

Q012 | WRITTEN

C07-K01-7-3-CH

A dropped stone has displacement $s(t) = 4.9t^2$ metres. Find its instantaneous velocity at $t = 10$ seconds using the definition.

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F04 | V01

Q013 | WRITTEN

C07-K01-7-4-WU

Differentiate $f(x) = x^2 + 3x$ using the definition of the derivative.

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F04 | V02

Q014 | WRITTEN

C07-K01-7-4-TRY

Differentiate the three given Try functions using the definition.

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F04 | V03

Q015 | WRITTEN

C07-K01-7-4-EX

Differentiate all nine given Exercise functions using the definition.

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F04 | V04

Q016 | WRITTEN

C07-K01-7-4-CH

A regular hexagon has side length x . Write its area and differentiate it using the definition.

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F04 | V05

Q017 | WRITTEN

C07-K01-7-4-REF

If $f(x) = 5$ for every x except that it is undefined at $x=2$, does $\lim_{x \rightarrow 2} f(x)$ exist?

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F01 | C07-K01-F01-VMCQ

Q018 | MCQ

C07-K01-7-1-GM01

The average rate of change from a to b is:

A $[f(a)+f(b)]/(b-a)$

B $[f(b)-f(a)]/(b-a)$

C $f'(a)$

D $b-a$

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F01 | C07-K01-F01-VMCQ

Q019 | MCQ

C07-K01-7-1-GM02

For $f(x) = x^2 + 2x$ from -1 to 3, the average rate is:

A 2

B 3

C 4

D 8

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2} x^2 \quad A' = 3\sqrt{3} x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F01 | C07-K01-F01-VMCQ

Q020 | MCQ

C07-K01-7-1-GM03

The average area-rate for diameter 10 to 18 is:

A 7π B 14π C 28π D 56π

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F01 | C07-K01-F01-VMCQ

Q021 | MCQ

C07-K01-7-1-GM04

The h-form average rate is found by Which of the following is correct?:

A setting $h=1$ B cancelling the common h

C differentiating first

D squaring h

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F02 | C07-K01-F02-VMCQ

Q022 | MCQ

C07-K01-7-2-GM05

A direct-substitution limit is valid when Which of the following is correct?:

A the expression is defined

B the denominator is zero

C the numerator is infinite

D h is negative

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F02 | C07-K01-F02-VMCQ

Q023 | MCQ

C07-K01-7-2-GM06

The 0/0 form requires Which of the following is correct?:

A factor/cancel first

B differentiate twice

C ignore the denominator

D change the base

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2} x^2 \quad A' = 3\sqrt{3} x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F02 | C07-K01-F02-VMCQ

Q024 | MCQ

C07-K01-7-2-GM07

In the finite-limit challenge, k equals Which of the following is correct?:

A 3

B 5

C 6

D 9

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F02 | C07-K01-F02-VMCQ

Q025 | MCQ

C07-K01-7-2-GM08

For a polynomial, the limit equals Which of the following is correct?:

☐ A its value at the point

☐ B zero

☐ C infinity

☐ D the slope

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F03 | C07-K01-F03-VMCQ

Q026 | MCQ

C07-K01-7-3-GM09

The definition of the differential coefficient at a is:

A $\lim [f(a+h)-f(a)]/h$

B $f(a+h)+f(a)$

C $[f(a)-f(0)]/a$

D $h/f(a)$

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F03 | C07-K01-F03-VMCQ

Q027 | MCQ

C07-K01-7-3-GM10

For $f(x) = 3x^2$, $f'(-2)$ is:

A -12

B -6

C 6

D 12

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F03 | C07-K01-F03-VMCQ

Q028 | MCQ

C07-K01-7-3-GM11

The stone's velocity at 10 s is:

A 49

B 98

C 196

D 4.9

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F03 | C07-K01-F03-VMCQ

Q029 | MCQ

C07-K01-7-3-GM12

After cancelling h, take the limit as Which of the following is correct?:

A $h \rightarrow 0$ B $h \rightarrow 1$ C $x \rightarrow 0$ D $a \rightarrow \infty$

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F04 | C07-K01-F04-VMCQ

Q030 | MCQ

C07-K01-7-4-GM13

The derivative function is:

A the coefficient at one point

B the differential coefficient for every x

C the average rate only

D the original function

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F04 | C07-K01-F04-VMCQ

Q031 | MCQ

C07-K01-7-4-GM14

The derivative of $x^2 + 3x$ from the definition is:A $x+3$ B $2x+3$ C $2x$ D x^2

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F04 | C07-K01-F04-VMCQ

Q032 | MCQ

C07-K01-7-4-GM15

The area of a regular hexagon of side x is:

A $3\sqrt{3}x$

B $(3\sqrt{3}/2)x^2$

C $6x^2$

D $\sqrt{3}x$

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F04 | C07-K01-F04-VMCQ

Q033 | MCQ

C07-K01-7-4-GM16

A function need not be defined at a for its limit to Which of the following is correct?:

A exist

B be infinite only

C equal zero only

D fail

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F04 | C07-K01-F04-VMCQ

Q034 | MCQ

C07-K01-7-4-GM17

The derivative of a constant function is:

A 0

B 1

C x

D the constant

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F01 | C07-K01-F01-VQUIZ

Quiz M1 | QUIZ MCQ

C07-K01-Q-M01

Find the average rate of change of x^2 from 1 to 3:

A 4

B 1

C 2

D 3

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F02 | C07-K01-F02-VQUIZ

Quiz M2 | QUIZ MCQ

C07-K01-Q-M02

Find $\lim_{x \rightarrow 2} (x^2 + 1)$:

A 0

B 5

C 2

D 3

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2} x^2 \quad A' = 3\sqrt{3} x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F03 | C07-K01-F03-VQUIZ

Quiz M3 | QUIZ MCQ

C07-K01-Q-M03

Find $f'(x)$ for $f(x) = x^2$ using the definition:

A 0

B 1

C $2x$

D 3

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F03 | C07-K01-F03-VQUIZ

Quiz M4 | QUIZ MCQ

C07-K01-Q-M04

If $s(t) = 4.9t^2$, find the velocity at $t=5$:

A 0

B 1

C 2

D 49

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F01 | C07-K01-F01-VQUIZ

Quiz W1 | QUIZ WRITTEN

C07-K01-Q-W01

Find the average rate of change of $3x + 1$ from 2 to 7.

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F02 | C07-K01-F02-VQUIZ

Quiz W2 | QUIZ WRITTEN

C07-K01-Q-W02

Evaluate $\lim_{h \rightarrow 0} (6 + 2h)$.

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F03 | C07-K01-F03-VQUIZ

Quiz W3 | QUIZ WRITTEN

C07-K01-Q-W03

Find the differential coefficient of $2x^2$ at $x=3$ from the definition.

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2}x^2 \quad A' = 3\sqrt{3}x$$

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

F04 | C07-K01-F04-VQUIZ

Quiz W4 | QUIZ WRITTEN

C07-K01-Q-W04

State $\lim_{x \rightarrow 2} f(x)$ when $f(x)=5$ for all x except $x=2$ where it is undefined.

WORKING AREA

FORMULA BANK

Average rate

$$\frac{f(b) - f(a)}{b - a}$$

Limit

$$\lim_{x \rightarrow a} f(x) = \alpha$$

Coefficient

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Hexagon area

$$A = \frac{3\sqrt{3}}{2} x^2 \quad A' = 3\sqrt{3}x$$

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 07

Differentiation and Integration

1 Concepts | 40 questions

B5 Landscape | one question per teaching page

Eng Eslam Ahmed

Assistant Lecturer • Faculty of Engineering • Cairo University

01120009622

Chapter 07 | Differentiation and Integration

CLASSROOM ROUTE

Concept 2 | Differentiation and Tangent Lines**40 questions**

Official lessons: 7-5, 7-6, 7-7, 7-8

The Formula Bank changes at each Concept boundary.

Concept 2 | Differentiation and Tangent Lines

OFFICIAL LESSONS 7-5, 7-6, 7-7, 7-8

Differentiation and Tangent Lines

Equation-first recall | use the same rail beside every question in this Concept

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Start with the family cue, write the first move, then use the working grid.

Concept 2 | Differentiation and Tangent Lines

F01 | V01

Q001 | WRITTEN

C07-K02-7-5-WU

Differentiate the four Warm Up demands: $y = x^2 - 3x + 2$, $y = (2x + 1)^3$, $V = \pi r^2 h$ with respect to (r) , and find $f'(-1)$ for $f(x) = x^3 - 3x + 1$.

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F01 | V02

Q002 | WRITTEN

C07-K02-7-5-TRY

Differentiate the six printed Try functions, the two variable-labelled functions $S = \pi r^2$ and $h = vt - \frac{1}{2}gt^2$, and evaluate $f'(2)$, $f'(-3)$ for $f(x) = 4x^3 - 3x^2 + 2$.

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F01 | V03

Q003 | WRITTEN

C07-K02-7-5-EX

Differentiate every given Exercise function 18 *polynomial/product forms*, the six functions with variables indicated in brackets, and evaluate the two values in each of Exercises 3–5.

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F01 | V04

Q004 | WRITTEN

C07-K02-7-5-CH

For $f(x) = \sqrt[3]{x^5} + \frac{1}{x^2}$, find $f'(1) + f'(-1)$.

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F02 | V01

Q005 | WRITTEN

C07-K02-7-6-WU

Find the quadratic $f(x) = ax^2 + bx + c$ satisfying $f'(0) = -2$, $f'(2) = 2$, $f(1) = 2$.

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F02 | V02

Q006 | WRITTEN

C07-K02-7-6-TRY

Find the quadratic satisfying $f'(0) = 2$, $f'(1) = 4$, $f(2) = 6$.

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F02 | V03

Q007 | WRITTEN

C07-K02-7-6-EX

Find the four quadratics satisfying the four given triples of derivative and function conditions.

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F02 | V04

Q008 | WRITTEN

C07-K02-7-6-CH

Find the quadratic $f(x)$ satisfying $f'(2) + f'(4) = 0$, $f(3) = 0$, $f(1) = 12$.

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F03 | V01

Q009 | WRITTEN

C07-K02-7-7-WU

Find the tangent line to $y = x^2 - 4x + 1$ at $(3, -2)$.

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F03 | V02

Q010 | WRITTEN

C07-K02-7-7-TRY

Find the tangent line for the three given functions at their stated points: $y = x^2 + 3$ at $(-2, 7)$, $y = x^2 - 3x + 2$ at $((1, 0))$, and $y = -x^3 + 2x + 4$ at $(-2, 8)$.

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F03 | V03

Q011 | WRITTEN

C07-K02-7-7-EX

Find the tangent line for all nine given Exercise functions at their given points.

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

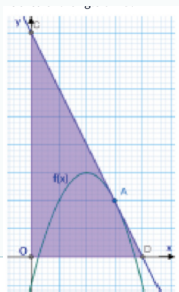
Concept 2 | Differentiation and Tangent Lines

F03 | V04

Q012 | WRITTEN

C07-K02-7-7-CH

For $f(x) = -x^2 + 4x - 1$, the tangent at x-coordinate 3 meets the axes at (D, C). Find the area of triangle (ODC).



WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F04 | V01

Q013 | WRITTEN

C07-K02-7-8-WU

Solve the two given Warm Up demands: tangents from $(C(0, 2))$ to $y = x^2 - 3x + 6$, and tangents to $y = x^3 + 4x^2 - 3$ with gradient 3.

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F04 | V02

Q014 | WRITTEN

C07-K02-7-8-TRY

Solve the two given Try demands: tangents from $C(1, -1)$ to $y = x^2 - x + 3$, and tangents to $y = x^3 + 2x - 3$ with gradient 5.

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F04 | V03

Q015 | WRITTEN

C07-K02-7-8-EX

Solve all six given Exercise demands involving external points or prescribed gradients.

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

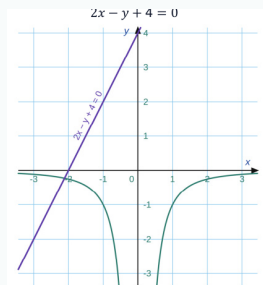
Concept 2 | Differentiation and Tangent Lines

F04 | V04

Q016 | WRITTEN

C07-K02-7-8-CH

For $f(x) = -1/x^2$, find the tangent line parallel to $2x - y + 4 = 0$.



WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F01 | C07-K02-F01-VMCQ

Q017 | MCQ

C07-K02-7-5-GM01

The derivative of x^n is:

A x^{n-1}

B nx^{n-1}

C $n + x$

D x^n

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F01 | C07-K02-F01-VMCQ

Q018 | MCQ

C07-K02-7-5-GM02

The derivative of $V = \pi r^2 h$ with respect to r is:

A πh B $2\pi r h$ C $2\pi r$ D πr^2

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F01 | C07-K02-F01-VMCQ

Q019 | MCQ

C07-K02-7-5-GM03

For $f(x) = x^3 - 3x + 1$, $f'(-1)$ is:

A -6

B -3

C 0

D 3

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F01 | C07-K02-F01-VMCQ

Q020 | MCQ

C07-K02-7-5-GM04

The challenge function can be written as Which of the following is correct?:

A $x^{5/3} + x^{-2}$

B $x^5 + x^2$

C $x^{3/5} + x^2$

D $x^{5/3} + x^2$

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F02 | C07-K02-F02-VMCQ

Q021 | MCQ

C07-K02-7-6-GM01

For $f(x) = ax^2 + bx + c$, $(f'(x))$ is:

A $ax + b$

B $2ax + b$

C $2a + b + c$

D $ax^2 + b$

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F02 | C07-K02-F02-VMCQ

Q022 | MCQ

C07-K02-7-6-GM02

The condition $f'(0) = -2$ gives Which of the following is correct?:

A $a = -2$

B $b = -2$

C $c = -2$

D $2a = -2$

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F02 | C07-K02-F02-VMCQ

Q023 | MCQ

C07-K02-7-6-GM03

The warm-up quadratic is:

A $x^2 - 2x + 3$

B $x^2 + 2x + 3$

C $2x^2 - x + 3$

D $x^2 - 2x - 3$

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F02 | C07-K02-F02-VMCQ

Q024 | MCQ

C07-K02-7-6-GM04

The challenge coefficient a is:

A 1

B 2

C 3

D 6

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F03 | C07-K02-F03-VMCQ

Q025 | MCQ

C07-K02-7-7-GM01

The tangent gradient at $x=a$ is:A $(f(a))$ B (a) C $(f'(a))$ D $1/f'(a)$

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F03 | C07-K02-F03-VMCQ

Q026 | MCQ

C07-K02-7-7-GM02

The tangent to $y = x^2 - 4x + 1$ at $(3, -2)$ is:

A $y = x - 5$

B $y = 2x - 8$

C $y = -2x + 4$

D $y = 3x - 11$

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F03 | C07-K02-F03-VMCQ

Q027 | MCQ

C07-K02-7-7-GM03

In the triangle challenge, the x-intercept of the tangent is:

A 2

B 3

C 4

D 8

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F03 | C07-K02-F03-VMCQ

Q028 | MCQ

C07-K02-7-7-GM04

The area of triangle (ODC) in the challenge is:

A 8

B 12

C 16

D 32

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F04 | C07-K02-F04-VMCQ

Q029 | MCQ

C07-K02-7-8-GM01

When the contact point is not given, set it to Which of the following is correct?:

A $((0,0))$ B $((a,f(a)))$ C $((f(a),a))$ D $((1,f'(a)))$

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F04 | C07-K02-F04-VMCQ

Q030 | MCQ

C07-K02-7-8-GM02

The gradient of $2x - y + 4 = 0$ is:

A -2

B -1

C 1

D 2

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F04 | C07-K02-F04-VMCQ

Q031 | MCQ

C07-K02-7-8-GM03

For $f(x) = -1/x^2$, $(f'(x))$ is:

A $-2/x^3$

B $2/x^3$

C $1/x^2$

D $-1/x^3$

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F04 | C07-K02-F04-VMCQ

Q032 | MCQ

C07-K02-7-8-GM04

The parallel tangent in the challenge is:

A $y = 2x + 3$

B $y = 2x - 3$

C $y = -2x - 3$

D $y = x - 2$

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F01 | C07-K02-F01-VQUIZ

Quiz W1 | QUIZ WRITTEN

C07-K02-Q-M01

Differentiate $3x^3 - 2x$.

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F02 | C07-K02-F02-VQUIZ

Quiz W2 | QUIZ WRITTEN

C07-K02-Q-M02

Find the quadratic with $f'(0) = 1$, $f'(1) = 3$, $f(1) = 2$.

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F03 | C07-K02-F03-VQUIZ

Quiz W3 | QUIZ WRITTEN

C07-K02-Q-M03

Find the tangent to $y = x^2$ at $((2, 4))$.

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F04 | C07-K02-F04-VQUIZ

Quiz W4 | QUIZ WRITTEN

C07-K02-Q-M04

For $f(x) = -1/x$, find the tangent parallel to $y = x + 2$.

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F01 | C07-K02-F01-VQUIZ

Quiz M1 | QUIZ MCQ

C07-K02-Q-M05

The derivative of a constant is:

A 0

B 1

C x

D the constant

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F02 | C07-K02-F02-VQUIZ

Quiz M2 | QUIZ MCQ

C07-K02-Q-M06

For $f(x) = ax^2 + bx + c$, the coefficient equations come from:

A three function values

B two derivatives and one function value

C one derivative only

D the tangent gradient only

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F03 | C07-K02-F03-VQUIZ

Quiz M3 | QUIZ MCQ

C07-K02-Q-M07

The tangent equation in point-slope form is:

A $y - y_1 = m(x - x_1)$

B $y = mx + c$

C $y + y_1 = m(x + x_1)$

D $x - x_1 = m(y - y_1)$

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

Concept 2 | Differentiation and Tangent Lines

F04 | C07-K02-F04-VQUIZ

Quiz M4 | QUIZ MCQ

C07-K02-Q-M08

A tangent parallel to $y = 3x - 1$ has gradient Which of the following is correct?:

A -3

B $1/3$

C 3

D 1

WORKING AREA

FORMULA BANK

Power rule

$$(x^n)' = nx^{n-1}$$

Quadratic

$$f(x) = ax^2 + bx + c \quad f'(x) = 2ax + b$$

Tangent

$$y - y_1 = m(x - x_1) \quad m = f'(a)$$

Unknown contact

$$(a, f(a)) \quad y - f(a) = f'(a)(x - a)$$

Parallel lines

$$m_1 = m_2$$

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 07

Differentiation and Integration

1 Concepts | 76 questions

B5 Landscape | one question per teaching page

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Chapter 07 | Differentiation and Integration

CLASSROOM ROUTE

Concept 3 | Applications of Differentiation: Polynomial Functions**76 questions**

Official lessons: 7-9, 7-10, 7-11, 7-12, 7-13, 7-14, 7-15, 7-16, 7-17

The Formula Bank changes at each Concept boundary.

Concept 3 | Applications of Differentiation: Polynomial Functions

OFFICIAL LESSONS 7-9, 7-10, 7-11, 7-12, 7-13, 7-14, 7-15, 7-16, 7-17

Applications of Differentiation: Polynomial Functions

Equation-first recall | use the same rail beside every question in this Concept

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Start with the family cue, write the first move, then use the working grid.

Concept 3 | Applications of Differentiation: Polynomial Functions

F01 | V01

Q001 | WRITTEN

C07-K03-7-9-WU

Find the extreme values and sketch the graphs of the three cubic functions printed in the Warm Up.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F01 | V02

Q002 | WRITTEN

C07-K03-7-9-TRY

Find the extreme values and sketch the three given Try cubics.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F01 | V03

Q003 | WRITTEN

C07-K03-7-9-EX

Find the extreme values and draw the graphs for all nine given cubic functions in Exercise.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F01 | V04

Q004 | WRITTEN

C07-K03-7-9-CH

For $P(x) = x^3 - 3ax^2 + (3a^2 - 3)x + 4$, $a > 0$, find the flow rates giving the maximum and minimum peak efficiency.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F02 | V01

Q005 | WRITTEN

C07-K03-7-10-WU

On $[-3, 4]$, find the maximum and minimum values of $y = -x^3 + 12x + 5$.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F02 | V02

Q006 | WRITTEN

C07-K03-7-10-TRY

Find the maximum and minimum values of the two given Try cubics on their stated closed intervals.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F02 | V03

Q007 | WRITTEN

C07-K03-7-10-EX

Find the maximum and minimum values of all six given cubics on their given closed intervals.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F02 | V04

Q008 | WRITTEN

C07-K03-7-10-CH

For $f(x) = x^3 - 6x$ on $[-3, 3]$, find the absolute deepest dive and highest ascent.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F03 | V01

Q009 | WRITTEN

C07-K03-7-11-WU

If $f(x) = -x^3 + ax + b$ has a local maximum of 9 at $x = 2$, find a , b , and the local minimum.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F03 | V02

Q010 | WRITTEN

C07-K03-7-11-TRY

Solve the two given Try problems: determine the coefficients from a local minimum/maximum and another stationary point.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F03 | V03

Q011 | WRITTEN

C07-K03-7-11-EX

Solve all six given Exercise problems asking for cubic coefficients from extrema.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F03 | V04

Q012 | WRITTEN

C07-K03-7-11-CH

For $f(x) = x^3 + ax^2 + bx + c$, the graph has a local peak 178 at $x = -2$ and a local trough at $x = 4$. Find a , b , c .

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F04 | V01

Q013 | WRITTEN

C07-K03-7-12-WU

If $f(x) = x^3 - 3kx^2 + kx + 2$ has no extreme values, find the range of k .

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F04 | V02

Q014 | WRITTEN

C07-K03-7-12-TRY

Determine the parameter ranges in the given Try problems for a cubic to have no extreme values or to have both extremes.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F04 | V03

Q015 | WRITTEN

C07-K03-7-12-EX

Determine the parameter ranges for all six given Exercise cubics according to whether they have extreme values.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F04 | V04

Q016 | WRITTEN

C07-K03-7-12-CH

For $f(x) = x^3 + 3kx^2 + 12kx + 10$, find the exact k -range that guarantees both a local maximum and minimum.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F05 | V01

Q017 | WRITTEN

C07-K03-7-13-WU

If $f(x) = x^3 - 3kx^2 + (k^2 + 5)x + 3$ is always monotonically increasing, find the range of k .

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F05 | V02

Q018 | WRITTEN

C07-K03-7-13-TRY

Find the parameter ranges in the two given Try problems for the cubic to be monotonically increasing.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F05 | V03

Q019 | WRITTEN

C07-K03-7-13-EX

Find the parameter ranges in all six given Exercise problems for monotonic increase.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F05 | V04

Q020 | WRITTEN

C07-K03-7-13-CH

For $f(x) = \frac{1}{3}x^3 - kx^2 + (k+6)x + 10$, find the k -range for monotonic increase on $x \in [1, \infty)$.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | V01

Q021 | WRITTEN

C07-K03-7-14-WU

Find the number of distinct real roots of $x^3 - 6x^2 + 9x - 3 = 0$.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | V02

Q022 | WRITTEN

C07-K03-7-14-TRY

Find the number of distinct real roots for the three given Try equations.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | V03

Q023 | WRITTEN

C07-K03-7-14-EX

Find the number of distinct real roots for all nine given cubic equations.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | V04

Q024 | WRITTEN

C07-K03-7-14-CH

Find the number of distinct real roots of $-h^3 + 12h^2 - 36h + 40 = 0$.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | V05

Q025 | WRITTEN

C07-K03-7-15-WU

If $-x^3 + 3x^2 + 9x + a = 0$ has three distinct real roots, find the range of a .

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | V06

Q026 | WRITTEN

C07-K03-7-15-TRY

Solve the two given Try parameter problems for three or two distinct real roots.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | V07

Q027 | WRITTEN

C07-K03-7-15-EX

Solve all four given Exercise parameter problems for the required number of distinct roots.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | V08

Q028 | WRITTEN

C07-K03-7-15-CH

For $f(x) = x^3 - 6x^2 + 9x - 2a^2 + 7a$, find all real a such that $f(x) = 2$ has exactly three distinct real roots.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | V09

Q029 | WRITTEN

C07-K03-7-16-WU

Prove the given inequalities using an auxiliary cubic and a derivative sign table.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | V10

Q030 | WRITTEN

C07-K03-7-16-EX

Prove all three given cubic inequalities on their stated domains.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | V11

Q031 | WRITTEN

C07-K03-7-16-CH

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ for every $x \in [-1, 3]$.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | V12

Q032 | WRITTEN

C07-K03-7-17-WU

Find the extreme values and sketch the two given quartic functions.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | V13

Q033 | WRITTEN

C07-K03-7-17-EX

Find the extreme values and graphs for all six given quartic functions.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | V14

Q034 | WRITTEN

C07-K03-7-17-REF

For $f(x) = x^4 - 4x^3 + k$, find the exact k-range for exactly two distinct real roots.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F01 | C07-K03-F01-VMCQ

Q035 | MCQ

C07-K03-7-9-GM01

At a local maximum, the sign of f' changes Which of the following is correct?:

A $- \rightarrow +$ B $+ \rightarrow -$ C $+ \rightarrow +$ D $- \rightarrow -$

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F01 | C07-K03-F01-VMCQ

Q036 | MCQ

C07-K03-7-9-GM02

For $P'(x) = 3[(x - a)^2 - 1]$, the local maximum occurs at Which of the following is correct?:

A $a - 1$ B a C $a + 1$ D $1 - a$

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F02 | C07-K03-F02-VMCQ

Q037 | MCQ

C07-K03-7-10-GM03

Absolute extrema on a closed interval require comparing Which of the following is correct?:

A stationary points only

B endpoints only

C stationary points and endpoints

D the y-intercept

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F02 | C07-K03-F02-VMCQ

Q038 | MCQ

C07-K03-7-10-GM04

The absolute maximum of $x^3 - 6x$ on $[-3, 3]$ is:

A $4\sqrt{2}$

B 9

C 6

D 0

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F03 | C07-K03-F03-VMCQ

Q039 | MCQ

C07-K03-7-11-GM05

If $f(x) = -x^3 + ax + b$ has a stationary point at 2, then a is:

A 2

B 6

C 12

D -12

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F03 | C07-K03-F03-VMCQ

Q040 | MCQ

C07-K03-7-11-GM06

The derivative roots in the challenge are:

A -4, 2

B -2, 4

C 2, 4

D -3, 3

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F04 | C07-K03-F04-VMCQ

Q041 | MCQ

C07-K03-7-12-GM07

No two distinct extrema for a cubic means the derivative discriminant is:

A $D > 0$

B $D = 1$

C $D \leq 0$

D $D < 0$ only

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F04 | C07-K03-F04-VMCQ

Q042 | MCQ

C07-K03-7-12-GM08

The challenge range for both extrema is:

A $0 < k < 4$

B $k < 0$ or $k > 4$

C $-4 < k < 0$

D $k \leq 4$

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F05 | C07-K03-F05-VMCQ

Q043 | MCQ

C07-K03-7-13-GM09

For a positive-leading cubic to be monotone increasing, its derivative must be Which of the following is correct?:

A negative everywhere

B non-negative everywhere

C zero everywhere

D linear

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F05 | C07-K03-F05-VMCQ

Q044 | MCQ

C07-K03-7-13-GM10

The challenge domain in 7-13 is:

A $(-\infty, \infty)$ B $[1, \infty)$ C $[-1, 1]$ D $(0, 1)$

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | C07-K03-F06-VMCQ

Q045 | MCQ

C07-K03-7-14-GM11

Three distinct roots occur when the x-axis lies Which of the following is correct?:

☐ A above both extrema☐ B between the local maximum and minimum☐ C at an endpoint only☐ D at the local maximum only

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | C07-K03-F06-VMCQ

Q046 | MCQ

C07-K03-7-14-GM12

The 7-14 challenge has how many distinct real roots? Which of the following is correct?:

A 0

B 1

C 2

D 3

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | C07-K03-F06-VMCQ

Q047 | MCQ

C07-K03-7-15-GM13

Three intersections with $y = a$ require the level to lie Which of the following is correct?:

☐ A strictly between the extrema☐ B above the maximum☐ C below the minimum☐ D at an extremum

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | C07-K03-F06-VMCQ

Q048 | MCQ

C07-K03-7-15-GM14

The 7-15 warm-up range is:

A $-27 < a < 5$

B $-5 < a < 27$

C $a \leq -27$

D $a \geq 5$

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | C07-K03-F06-VMCQ

Q049 | MCQ

C07-K03-7-16-GM15

To prove $LHS \geq RHS$, define Which of the following is correct?:

A $f = LHS + RHS$

B $f = LHS - RHS$

C $f = RHS - 1$

D $f = f'$

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | C07-K03-F06-VMCQ

Q050 | MCQ

C07-K03-7-16-GM16

The largest k in the challenge is:

A $2 - 4\sqrt{2}$

B $4\sqrt{2} - 2$

C $2 + 4\sqrt{2}$

D $-2 - 4\sqrt{2}$

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | C07-K03-F06-VMCQ

Q051 | MCQ

C07-K03-7-17-GM17

A repeated root of a derivative may be Which of the following is correct?:

A always a local maximum

B always a local minimum

C stationary but not an extreme

D impossible

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | C07-K03-F06-VMCQ

Q052 | MCQ

C07-K03-7-17-GM18

Exactly two distinct roots for $x^4 - 4x^3 + k = 0$ require Which of the following is correct?:

A $k = 27$ B $k < 27$ C $k > 27$ D $k \leq 0$ only

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F01 | C07-K03-F01-VMCQ

Q053 | MCQ

C07-K03-7-9-GM19

A local minimum is identified by the change Which of the following is correct?:

A $+$ $\rightarrow$ $-$ B $-$ $\rightarrow$ $+$ C $+$ $\rightarrow$ $+$ D $-$ $\rightarrow$ $-$

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F01 | C07-K03-F01-VMCQ

Q054 | MCQ

C07-K03-7-9-GM20

For $y = x^3 - 6x^2 + 16$, the critical x-values are:

A 0, 4

B 1, 3

C -2, 2

D 2, 4

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F01 | C07-K03-F01-VMCQ

Q055 | MCQ

C07-K03-7-9-GM21

The derivative sign table is used to determine:

A only the y-intercept

B increase/decrease and extrema

C the constant term

D the domain

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F02 | C07-K03-F02-VMCQ

Q056 | MCQ

C07-K03-7-10-GM22

On a closed interval, an endpoint can be Which of the following is correct?:

☐ A ignored☐ B an absolute extreme☐ C a derivative root only☐ D never a minimum

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F02 | C07-K03-F02-VMCQ

Q057 | MCQ

C07-K03-7-10-GM23

For $y = -x^3 + 12x + 5$, the stationary x-values are:

A -2, 2

B -1, 1

C 0, 4

D 2, 4

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F02 | C07-K03-F02-VMCQ

Q058 | MCQ

C07-K03-7-10-GM24

The absolute minimum of $x^3 - 6x$ on $[-3, 3]$ is at Which of the following is correct?:

A $x = -3$

B $x = -\sqrt{2}$

C $x = \sqrt{2}$

D $x = 3$

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F03 | C07-K03-F03-VMCQ

Q059 | MCQ

C07-K03-7-11-GM25

To determine a cubic from an extreme value, use both $f(a) = M$ and:

A $f(a) = 0$

B $f'(a) = 0$

C $f''(a) = 0$

D $a = 0$

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F03 | C07-K03-F03-VMCQ

Q060 | MCQ

C07-K03-7-11-GM26

In the warm-up $f(x) = -x^3 + 12x - 7$, the local minimum occurs at Which of the following is correct?:

A $x = -2$

B $x = 0$

C $x = 2$

D $x = 7$

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F04 | C07-K03-F04-VMCQ

Q061 | MCQ

C07-K03-7-12-GM27

For $f(x) = x^3 - 3kx^2 + kx + 2$, no two distinct stationary roots requires Which of the following is correct?:

A $0 \leq k \leq 1/3$

B $k > 1/3$

C $k < 0$

D $k \geq 1/3$

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F04 | C07-K03-F04-VMCQ

Q062 | MCQ

C07-K03-7-12-GM28

Two distinct stationary points occur when the derivative discriminant is:

A $D < 0$

B $D = 0$

C $D > 0$

D $D = 1$

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F05 | C07-K03-F05-VMCQ

Q063 | MCQ

C07-K03-7-13-GM29

For a positive-leading quadratic derivative, $D \leq 0$ implies it is:

A non-negative for all x

B negative for all x

C linear

D undefined

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F05 | C07-K03-F05-VMCQ

Q064 | MCQ

C07-K03-7-13-GM30

In the 7-13 challenge, monotonicity is required on Which of the following is correct?:

A $[1, \infty)$ B $[-1, 1]$ C $(-\infty, 0)$ D $[0, 1]$

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | C07-K03-F06-VMCQ

Q065 | MCQ

C07-K03-7-14-GM31

A cubic with both stationary values above the x-axis has Which of the following is correct?:

A three roots

B one distinct real root

C no real roots

D two repeated roots

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | C07-K03-F06-VMCQ

Q066 | MCQ

C07-K03-7-14-GM32

To count cubic roots, compare the extrema with Which of the following is correct?:

☐ A the y-axis☐ B the x-axis☐ C the derivative axis☐ D the domain only

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | C07-K03-F06-VMCQ

Q067 | MCQ

C07-K03-7-15-GM33

At an extremal parameter level, a cubic and a horizontal line have Which of the following is correct?:

A one distinct intersection

B two distinct intersections

C three distinct intersections

D no intersection

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | C07-K03-F06-VMCQ

Q068 | MCQ

C07-K03-7-17-GM34

For $x^4 - 4x^3 + k$, the global minimum occurs at Which of the following is correct?:

A $x = 0$ B $x = 1$ C $x = 3$ D $x = 4$

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F01 | C07-K03-F01-VQUIZ

Quiz W1 | QUIZ WRITTEN

C07-K03-Q-W01

Find the local maximum and minimum of $x^3 - 3x^2 + 1$.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F03 | C07-K03-F03-VQUIZ

Quiz W2 | QUIZ WRITTEN

C07-K03-Q-W02

Find a, b if $f(x) = -x^3 + ax + b$ has a local maximum 4 at $x=1$.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F04 | C07-K03-F04-VQUIZ

Quiz W3 | QUIZ WRITTEN

C07-K03-Q-W03

Find the k-range for $x^3 - 3kx^2 + kx$ to have no two distinct stationary points.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | C07-K03-F06-VQUIZ

Quiz W4 | QUIZ WRITTEN

C07-K03-Q-W04

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ on $[-1, 3]$.

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F01 | C07-K03-F01-VQUIZ

Quiz M1 | QUIZ MCQ

C07-K03-Q-M05

At a local minimum, f' changes Which of the following is correct?:

A $+$ $\rightarrow$ $-$ B $-$ $\rightarrow$ $+$ C $+$ $\rightarrow$ $+$ D $-$ $\rightarrow$ $-$

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F05 | C07-K03-F05-VQUIZ

Quiz M2 | QUIZ MCQ

C07-K03-Q-M06

A cubic with a positive-leading derivative and $D < 0$ is:

☐ A always increasing☐ B always decreasing☐ C constant☐ D undefined

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | C07-K03-F06-VQUIZ

Quiz M3 | QUIZ MCQ

C07-K03-Q-M07

The 7-15 challenge requires the level to satisfy Which of the following is correct?:

A $t < 0$

B $0 < t < 4$

C $t > 4$

D $t = 4$

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

Concept 3 | Applications of Differentiation: Polynomial Functions

F06 | C07-K03-F06-VQUIZ

Quiz M4 | QUIZ MCQ

C07-K03-Q-M08

The global minimum of $x^4 - 4x^3 + k$ is:A $k - 27$ B $k + 27$ C $k - 3$ D $27 - k$

WORKING AREA

FORMULA BANK

First derivative

$$f'(x) > 0 \Rightarrow \nearrow$$

Stationary point

$$f'(a) = 0$$

Discriminant

$$D = b^2 - 4ac$$

Cubic roots

$$f(x) = 0 \Leftrightarrow \text{x-axis intersections}$$

Quartic

$$f'(x) = 0$$

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 07

Differentiation and Integration

1 Concepts | 58 questions

B5 Landscape | one question per teaching page

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Chapter 07 | Differentiation and Integration

CLASSROOM ROUTE

Concept 4 | Integration: Definite Integrals and Applications**58 questions**

Official lessons: 7-18, 7-19, 7-20, 7-21, 7-22, 7-23, 7-24

The Formula Bank changes at each Concept boundary.

Concept 4 | Integration: Definite Integrals and Applications

OFFICIAL LESSONS 7-18, 7-19, 7-20, 7-21, 7-22, 7-23, 7-24

Integration: Definite Integrals and Applications

Equation-first recall | use the same rail beside every question in this Concept

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Start with the family cue, write the first move, then use the working grid.

Concept 4 | Integration: Definite Integrals and Applications

F01 | V01

Q001 | WRITTEN

C07-K04-7-18-WU

Find the two indefinite integrals in the Warm Up.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F01 | V02

Q002 | WRITTEN

C07-K04-7-18-TRY

Find the six indefinite integrals in the given Try section, expanding a squared bracket first when needed.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F01 | V03

Q003 | WRITTEN

C07-K04-7-18-EX-A

Solve the first ten given indefinite-integral exercises (constants, linear and quadratic polynomials).

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F01 | V04

Q004 | WRITTEN

C07-K04-7-18-EX-B

Solve the remaining eight given integrals, including products and squared brackets.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F02 | V01

Q005 | WRITTEN

C07-K04-7-19-WU

Find $F(x)$ when $F'(x) = 3x^2 + 8x - 1$ and $F(1) = 7$.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F02 | V02

Q006 | WRITTEN

C07-K04-7-19-TRY

Find the two given Try functions from their derivatives and one given value of F .

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F02 | V03

Q007 | WRITTEN

C07-K04-7-19-EX

Find $F(x)$ for all six given Exercise conditions.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F03 | V01

Q008 | WRITTEN

C07-K04-7-20-WU

Find the curve $y = f(x)$ through $(3, 2)$ whose tangent gradient is $3x^2 - 2x + 1$.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F03 | V02

Q009 | WRITTEN

C07-K04-7-20-TRY

Solve the given Try tangent-gradient problem through (1, 1).

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F03 | V03

Q010 | WRITTEN

C07-K04-7-20-EX

Solve the three given tangent-gradient exercises, finding each function from its point and gradient.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F04 | V01

Q011 | WRITTEN

C07-K04-7-21-WU

Evaluate the three definite integrals in the Warm Up using the Fundamental Theorem of Calculus.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F04 | V02

Q012 | WRITTEN

C07-K04-7-21-TRY

Evaluate the six given Try definite integrals, simplifying the integrand before applying the limits.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F04 | V03

Q013 | WRITTEN

C07-K04-7-21-EX-A

Evaluate the first twelve given Exercise definite integrals, including reversed and equal limits.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F04 | V04

Q014 | WRITTEN

C07-K04-7-21-EX-B

Evaluate the remaining twelve given Exercise integrals, including sums, differences and products.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F05 | V01

Q015 | WRITTEN

C07-K04-7-22-WU

Use the four properties of definite integrals to simplify the three Warm Up expressions.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F05 | V02

Q016 | WRITTEN

C07-K04-7-22-TRY

Solve the four given Try definite-integral transformations.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F05 | V03

Q017 | WRITTEN

C07-K04-7-22-EX-A

Solve the first six given Exercise questions using interval splitting and reversal.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F05 | V04

Q018 | WRITTEN

C07-K04-7-22-EX-B

Solve the final six given Exercise questions using the factorized definite-integral identity.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F06 | V01

Q019 | WRITTEN

C07-K04-7-23-WU

Find $f(x)$ from the given equation $f(x) = x^2 + \int_0^1 f(t) dt$.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F06 | V02

Q020 | WRITTEN

C07-K04-7-23-TRY

Solve the two given Try integral equations for $f(x)$.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F06 | V03

Q021 | WRITTEN

C07-K04-7-23-EX

Find $f(x)$ for all five given integral equations.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F07 | V01

Q022 | WRITTEN

C07-K04-7-24-WU

Differentiate $\int_3^x (2t^2 + 3t - 5) dt$ with respect to x .

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F07 | V02

Q023 | WRITTEN

C07-K04-7-24-TRY

Solve the two given Try problems involving differentiation of a variable-limit integral and an unknown lower limit.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F07 | V03

Q024 | WRITTEN

C07-K04-7-24-EX-A

Differentiate the three given expressions with respect to x .

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F07 | V04

Q025 | WRITTEN

C07-K04-7-24-EX-B

Find $f(x)$ and a for the three given equations with a variable upper limit.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F01 | C07-K04-F01-VMCQ

Q026 | MCQ

C07-K04-7-18-GM01

The constant of integration is required because Which of the following is correct?:

A every primitive is unique

B primitives differ by a constant

C the integrand is constant

D the limits are equal

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F01 | C07-K04-F01-VMCQ

Q027 | MCQ

C07-K04-7-18-GM02

The integral of x^n for $n \neq -1$ is:

A $nx^{n-1} + C$

B $x^{n+1}/(n+1) + C$

C $x^n/n + C$

D $x^{n-1}/(n-1) + C$

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F01 | C07-K04-F01-VMCQ

Q028 | MCQ

C07-K04-7-18-GM03

Before integrating $(x - 1)^2$, first Which of the following is correct?:

A differentiate it

B expand it

C set $x = 0$

D add limits

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F01 | C07-K04-F01-VMCQ

Q029 | MCQ

C07-K04-7-18-GM04

The integral of $3x^2 - 2x + 1$ is:

A $x^3 - x^2 + x + C$

B $3x^3 - 2x^2 + x + C$

C $x^3 - 2x + 1 + C$

D $6x - 2 + C$

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F02 | C07-K04-F02-VMCQ

Q030 | MCQ

C07-K04-7-19-GM05

To find $F(x)$ from $F'(x)$, you should first:

A differentiate again

B integrate $F'(x)$

C set $F = 0$

D ignore the initial value

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F02 | C07-K04-F02-VMCQ

Q031 | MCQ

C07-K04-7-19-GM06

The condition $F(1) = 7$ determines Which of the following is correct?:

A the derivative

B the domain

C the constant C

D the leading coefficient

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F02 | C07-K04-F02-VMCQ

Q032 | MCQ

C07-K04-7-19-GM07

If $F'(x) = 6x - 2$ and $F(0) = 1$, then $F(x)$ is:

A $3x^2 - 2x + 1$

B $6x^2 - 2x + 1$

C $3x^2 - 2x$

D $6x - 2 + 1$

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F03 | C07-K04-F03-VMCQ

Q033 | MCQ

C07-K04-7-20-GM08

The gradient of the tangent at (x, y) is:A $f(x)$ B $f'(x)$ C $f''(x)$ D $x + y$

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F03 | C07-K04-F03-VMCQ

Q034 | MCQ

C07-K04-7-20-GM09

A point (x_0, y_0) on $y = f(x)$ gives Which of the following is correct?:

A $f'(x_0) = y_0$

B $f(x_0) = y_0$

C $f(0) = x_0$

D $C = 0$

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F03 | C07-K04-F03-VMCQ

Q035 | MCQ

C07-K04-7-20-GM10

If $f'(x) = 3x^2 - 2x + 1$, an antiderivative is:

A $3x^3 - 2x^2 + x$

B $x^3 - x^2 + x + C$

C $6x - 2$

D $x^3 - 2x + 1$

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F04 | C07-K04-F04-VMCQ

Q036 | MCQ

C07-K04-7-21-GM11

The Fundamental Theorem gives $\int_a^b f(x)dx =$ Which of the following is correct?:

A $F(a) - F(b)$

B $F(b) - F(a)$

C $F(a) + F(b)$

D 0 always

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F04 | C07-K04-F04-VMCQ

Q037 | MCQ

C07-K04-7-21-GM12

When the limits are equal, $\int_a^a f(x)dx$ equals Which of the following is correct?:

A 1

B a

C 0

D $f(a)$

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F04 | C07-K04-F04-VMCQ

Q038 | MCQ

C07-K04-7-21-GM13

A reversed definite integral satisfies Which of the following is correct?:

A $\int_a^b f = \int_b^a f$

B $\int_a^b f = -\int_b^a f$

C both are zero

D only if $f = 0$

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = -\int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F04 | C07-K04-F04-VMCQ

Q039 | MCQ

C07-K04-7-21-GM14

To evaluate a definite integral of a product, first:

A integrate each factor separately

B expand the product

C differentiate the product

D change the limits

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F05 | C07-K04-F05-VMCQ

Q040 | MCQ

C07-K04-7-22-GM15

Adjacent integrals with the same integrand can be Which of the following is correct?:

A multiplied

B combined over the joined interval

C set equal to zero

D reversed only

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F05 | C07-K04-F05-VMCQ

Q041 | MCQ

C07-K04-7-22-GM16

To use $\int_{\alpha}^{\beta} (x - \alpha)(x - \beta) dx$, the factors must have Which of the following is correct?:

A different roots

B the displayed roots (a,b)

C zero coefficients

D equal limits

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F05 | C07-K04-F05-VMCQ

Q042 | MCQ

C07-K04-7-22-GM17

The identity for the factored integral is:

A $(\beta - \alpha)^2/6$

B $-(\beta - \alpha)^3/6$

C $(\beta + \alpha)^3/6$

D 0

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F05 | C07-K04-F05-VMCQ

Q043 | MCQ

C07-K04-7-22-GM18

A minus sign before an integral can be removed by Which of the following is correct?:

A squaring the integrand

B reversing its limits

C adding C

D differentiating

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F06 | C07-K04-F06-VMCQ

Q044 | MCQ

C07-K04-7-23-GM19

In $f(x) = p(x) + \int_a^b f(t)dt$, the definite integral is:

A a function of x

B a constant

C always zero

D a derivative

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F06 | C07-K04-F06-VMCQ

Q045 | MCQ

C07-K04-7-23-GM20

The best first step for an integral equation is to let Which of the following is correct?:

A $A = f(x)$

B $A = \int_a^b f(t)dt$

C $A = x$

D $A = C$ only

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F06 | C07-K04-F06-VMCQ

Q046 | MCQ

C07-K04-7-23-GM21

For the warm-up $f(x) = x^2 + \int_0^1 f(t)dt$, the constant integral is:

A $1/3$ B $-1/3$ C 0 D 1

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F07 | C07-K04-F07-VMCQ

Q047 | MCQ

C07-K04-7-24-GM22

$\frac{d}{dx} \int_a^x f(t) dt$ equals Which of the following is correct?:

A $f(a)$ B $f(x)$ C $F(x)$

D 0

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F07 | C07-K04-F07-VMCQ

Q048 | MCQ

C07-K04-7-24-GM23

When differentiating an integral, the dummy variable t is replaced by Which of the following is correct?:

A the lower limit

B the upper-limit variable

C C

D zero

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F07 | C07-K04-F07-VMCQ

Q049 | MCQ

C07-K04-7-24-GM24

If $\int_a^a f(t)dt = 0$, this can determine:

A the derivative

B the unknown lower limit a

C the integrand degree

D the upper limit

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F07 | C07-K04-F07-VMCQ

Q050 | MCQ

C07-K04-7-24-GM25

For $\int_3^x (2t^2 + 3t - 5)dt$, the derivative is:

A $2x^2 + 3x - 5$

B $2x^3 + 3x^2 - 5x$

C $3x^2 + 3x - 5$

D 0

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F01 | C07-K04-F01-VQUIZ

Quiz W1 | QUIZ WRITTEN

C07-K04-Q-W01

Find $\int (4x^3 - 6x + 2)dx$.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F02 | C07-K04-F02-VQUIZ

Quiz W2 | QUIZ WRITTEN

C07-K04-Q-W02

Find $F(x)$ if $F'(x) = 2x - 5$ and $F(2) = 1$.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F04 | C07-K04-F04-VQUIZ

Quiz W3 | QUIZ WRITTEN

C07-K04-Q-W03

Evaluate $\int_{-1}^2 (3x^2 - 4x + 1)dx$.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F06 | C07-K04-F06-VQUIZ

Quiz W4 | QUIZ WRITTEN

C07-K04-Q-W04

Find $f(x)$ if $f(x) = x^2 + \int_0^1 f(t)dt$.

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F07 | C07-K04-F07-VQUIZ

Quiz M1 | QUIZ MCQ

C07-K04-Q-M05

The derivative of $\int_0^x (3t^2 + 1)dt$ is:A $3x^2 + 1$ B $x^3 + x$ C $6x$

D 0

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F01 | C07-K04-F01-VQUIZ

Quiz M2 | QUIZ MCQ

C07-K04-Q-M06

A primitive of $5x^4$ is:

A $x^5 + C$

B $5x^5 + C$

C $20x^3 + C$

D $x^4/5 + C$

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F02 | C07-K04-F02-VQUIZ

Quiz M3 | QUIZ MCQ

C07-K04-Q-M07

For $F'(x) = 4x + 2$ and $F(0) = 3$, $F(x)$ is:

A $2x^2 + 2x + 3$

B $4x^2 + 2x + 3$

C $2x^2 + 2$

D $4x + 5$

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

Concept 4 | Integration: Definite Integrals and Applications

F05 | C07-K04-F05-VQUIZ

Quiz M4 | QUIZ MCQ

C07-K04-Q-M08

$\int_1^3 f(x)dx + \int_3^5 f(x)dx$ equals Which of the following is correct?:

A $\int_1^5 f(x)dx$

B $\int_1^3 f(x)dx$

C $\int_3^5 f(x)dx$

D 0

WORKING AREA

FORMULA BANK

Power rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Linearity

$$\int (af + bg) dx = a \int f dx + b \int g dx$$

FTC

$$\int_a^b f(x) dx = F(b) - F(a)$$

Limit rules

$$\int_a^b f = - \int_b^a f$$

Variable limit

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 07

Differentiation and Integration

1 Concepts | 46 questions

B5 Landscape | one question per teaching page

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Chapter 07 | Differentiation and Integration

CLASSROOM ROUTE

Concept 5 | Applications of Integration to Geometry**46 questions**

Official lessons: 7-25, 7-26, 7-27, 7-28, 7-29, 7-30

The Formula Bank changes at each Concept boundary.

Concept 5 | Applications of Integration to Geometry

OFFICIAL LESSONS 7-25, 7-26, 7-27, 7-28, 7-29, 7-30

Applications of Integration to Geometry

Equation-first recall | use the same rail beside every question in this Concept

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Start with the family cue, write the first move, then use the working grid.

Concept 5 | Applications of Integration to Geometry

F01 | V01

Q001 | WRITTEN

C07-K05-7-25-WU-A

Find the area bounded by $y = 2x^2 + 1$, the x -axis, $x = -2$, and $x = 1$.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F01 | V02

Q002 | WRITTEN

C07-K05-7-25-WU-B

Find the area bounded by $y = x^2 + x - 6$, the x -axis, $x = -2$, and $x = 1$.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F01 | V03

Q003 | WRITTEN

C07-K05-7-25-WU-C

Find the area bounded by $y = -x^2 + 4$ and the x -axis.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F01 | V04

Q004 | WRITTEN

C07-K05-7-25-CH

Find the area under $y = -x^2 - 2x + 3$ and above the x -axis.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F02 | V01

Q005 | WRITTEN

C07-K05-7-26-WU-A

Find the area between $y = x^2 - 2$ and $y = -2x + 1$.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F02 | V02

Q006 | WRITTEN

C07-K05-7-26-WU-B

Find the area between $y = x^2 + 2x - 4$ and $y = -x^2 + 2x + 4$.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F02 | V03

Q007 | WRITTEN

C07-K05-7-26-WU-C

Find the area between $y = -x^2 + 4x$ and $y = x$.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F02 | V04

Q008 | WRITTEN

C07-K05-7-26-CH

Find the area between $y = -x^2 + 8$ and $y = 2x$ for $-2 \leq x \leq 3$.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F03 | V01

Q009 | WRITTEN

C07-K05-7-27-WU

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x -axis.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F03 | V02

Q010 | WRITTEN

C07-K05-7-27-CH

Find the area bounded by $y = -x^3 + 4x$ and the x -axis.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F03 | V03

Q011 | WRITTEN

C07-K05-7-27-EX

Solve the given cubic area exercises by finding every x -intercept and adding the absolute signed pieces.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F04 | V01

Q012 | WRITTEN

C07-K05-7-28-WU-A

Evaluate $\int_{-2}^1 |x + 1| \, dx$.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) \, dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) \, dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) \, dx = \frac{1}{2} \int_a^b f(x) \, dx$$

Concept 5 | Applications of Integration to Geometry

F04 | V02

Q013 | WRITTEN

C07-K05-7-28-WU-B

Evaluate $\int_0^3 |x^2 - 4| dx$.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F04 | V03

Q014 | WRITTEN

C07-K05-7-28-CH

Evaluate $\int_0^2 |x^2 + 3x - 4| dx$.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F05 | V01

Q015 | WRITTEN

C07-K05-7-29-WU-A

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F05 | V02

Q016 | WRITTEN

C07-K05-7-29-WU-B

Find the area bounded by $y = x^3 + x^2 - 3x + 6$ and its tangent at $(1, 5)$.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F05 | V03

Q017 | WRITTEN

C07-K05-7-29-CH

For $y = x^2$, tangents through $(3, 8)$ touch at $x = 2, 4$. Find the enclosed area.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F06 | V01

Q018 | WRITTEN

C07-K05-7-30-WU

If $y = kx$ bisects the area under $y = 2x - x^2$ and above the x -axis, find k .

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F06 | V02

Q019 | WRITTEN

C07-K05-7-30-EX

Solve the given bisection exercises for the given parabolas and lines, using the required area ratio.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F01 | C07-K05-F01-VMCQ

Q020 | MCQ

C07-K05-7-25-GM01

When the graph is below the x-axis, area is:

A the signed integral

B the negative of the integral

C always zero

D undefined

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F01 | C07-K05-F01-VMCQ

Q021 | MCQ

C07-K05-7-25-GM02

A bounded area under a curve is found by first checking Which of the following is correct?:

A the graph and sign

B *only they – intercept*

C the font size

D the derivative only

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F01 | C07-K05-F01-VMCQ

Q022 | MCQ

C07-K05-7-25-GM03

The area under $y = -x^2 + 4$ between its roots is:

A $8/3$ B $16/3$ C $32/3$ D $64/3$

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F02 | C07-K05-F02-VMCQ

Q023 | MCQ

C07-K05-7-26-GM04

Between two curves, the integrand is:

A *lower – upper*B *upper – lower*

C product

D derivative

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F02 | C07-K05-F02-VMCQ

Q024 | MCQ

C07-K05-7-26-GM05

If curves cross inside the interval, area must be Which of the following is correct?:

☐ A split at the crossing

☐ B ignored

☐ C doubled always

☐ D negative

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F02 | C07-K05-F02-VMCQ

Q025 | MCQ

C07-K05-7-26-GM06

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

A $16/3$ B $32/3$ C $64/3$ D $9/2$

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F03 | C07-K05-F03-VMCQ

Q026 | MCQ

C07-K05-7-27-GM07

For cubic area, the sign changes at Which of the following is correct?:

A every turning point

B *each* $x - \text{intercept}$

C *they* – *intercept* only

D the midpoint

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F03 | C07-K05-F03-VMCQ

Q027 | MCQ

C07-K05-7-27-GM08

The area for $y = -x^3 + 4x$ is:

A 4

B 6

C 8

D 12

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F03 | C07-K05-F03-VMCQ

Q028 | MCQ

C07-K05-7-27-GM09

A negative lobe contributes Which of the following is correct?:

☐ A negative area

☐ B positive absolute area

☐ C no area

☐ D a derivative

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F04 | C07-K05-F04-VMCQ

Q029 | MCQ

C07-K05-7-28-GM10

For an absolute-value integral, first locate:

A the roots

B the vertex only

C *they – axis*

D the tangent

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F04 | C07-K05-F04-VMCQ

Q030 | MCQ

C07-K05-7-28-GM11

The value of $\int_0^2 |x^2 + 3x - 4| dx$ is:

A 3

B 4

C 5

D 6

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F04 | C07-K05-F04-VMCQ

Q031 | MCQ

C07-K05-7-28-GM12

If f changes sign at c , split the integral Which of the following is correct?:

A at c

B at 0 only

C never

D after doubling

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F05 | C07-K05-F05-VMCQ

Q032 | MCQ

C07-K05-7-29-GM13

A tangent line at $x=a$ has slope Which of the following is correct?:

A $f(a)$ B $f'(a)$ C a^2

D the area

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F05 | C07-K05-F05-VMCQ

Q033 | MCQ

C07-K05-7-29-GM14

For two tangent lines, split at their Which of the following is correct?:

A *intersection* x – coordinate

B y – intercept only

C vertex only

D origin

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F05 | C07-K05-F05-VMCQ

Q034 | MCQ

C07-K05-7-29-GM15

The area in the given parabola-tangent Warm Up is:

A $1/2$

B 1

C $3/2$

D 3

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F06 | C07-K05-F06-VMCQ

Q035 | MCQ

C07-K05-7-30-GM16

Bisection means each part has Which of the following is correct?:

A the same area

B the same slope

C the same root

D zero area

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F06 | C07-K05-F06-VMCQ

Q036 | MCQ

C07-K05-7-30-GM17

For $y=2x-x^2$, the total area above the x-axis is:

A 1

B 2

C 3

D 4

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F06 | C07-K05-F06-VMCQ

Q037 | MCQ

C07-K05-7-30-GM18

The line $y=kx$ must satisfy the stated Which of the following is correct?:

A area ratio

B font ratio

C root count

D period

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F06 | C07-K05-F06-VMCQ

Q038 | MCQ

C07-K05-7-30-GM19

In a bisection problem, the cut-off area is:

A the total area

B half the total area

C twice the total area

D zero

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F01 | C07-K05-F01-VQUIZ

Quiz M1 | QUIZ MCQ

C07-K05-Q-M01

The area under $y = -x^2 + 4$ and above the x -axis is:A $16/3$ B 8 C $32/3$ D $64/3$

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F02 | C07-K05-F02-VQUIZ

Quiz M2 | QUIZ MCQ

C07-K05-Q-M02

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:A $16/3$ B $32/3$ C $64/3$ D $9/2$

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F04 | C07-K05-F04-VQUIZ

Quiz M3 | QUIZ MCQ

C07-K05-Q-M03

Evaluate $\int_{-2}^1 |x + 1| dx$:

A 2

B $5/2$

C 3

D $7/2$

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F05 | C07-K05-F05-VQUIZ

Quiz M4 | QUIZ MCQ

C07-K05-Q-M04

The area enclosed by $y = x^2$ and tangents through $(3, 8)$ is:

A $1/3$ B $1/2$ C $2/3$ D $4/3$

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F03 | C07-K05-F03-VQUIZ

Quiz W1 | QUIZ WRITTEN

C07-K05-Q-W01

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x -axis.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F04 | C07-K05-F04-VQUIZ

Quiz W2 | QUIZ WRITTEN

C07-K05-Q-W02

Evaluate $\int_0^2 |x^2 + 3x - 4| dx$.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F05 | C07-K05-F05-VQUIZ

Quiz W3 | QUIZ WRITTEN

C07-K05-Q-W03

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$

Concept 5 | Applications of Integration to Geometry

F06 | C07-K05-F06-VQUIZ

Quiz W4 | QUIZ WRITTEN

C07-K05-Q-W04

If $y = kx$ bisects the area under $y = 2x - x^2$, find k .

WORKING AREA

FORMULA BANK

Area under a curve

$$S = \int_a^b f(x) dx$$

Area between curves

$$S = \int_a^b (f(x) - g(x)) dx$$

Absolute value

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

Tangent line

$$y - y_1 = f'(x_1)(x - x_1)$$

Bisection of area

$$\int_a^k f(x) dx = \frac{1}{2} \int_a^b f(x) dx$$